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Comparison of methods for geologic storage of carbon dioxide in saline formations

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ABSTRACT

Preliminary estimates of CO_2 storage potential in geologic formations provide critical information related to Carbon Capture, Utilization, and Storage (CCUS) technologies to mitigate CO_2 emissions. Currently multiple methods to estimate CO_2 storage and multiple storage estimates for saline formations have been published, leading to potential uncertainty when comparing estimates from different studies. In this work, carbon dioxide storage estimates are compared by applying several commonly used methods to general saline formation data sets to assess the impact that the choice of method has on the results. Specifically, six CO_2 storage methods were applied to thirteen saline formation data sets which were based on formations across the United States with adaptations to provide the geologic inputs required by each method. Methods applied include those by (1) international efforts – the Carbon Sequestration Leadership Forum (Bachu et al., 2007); (2) United States government agencies – U.S. Department of Energy – National Energy Technology Laboratory (US-DOE-NETL, 2012) and United States Geological Survey (Brennan et al., 2010); and (3) the peer-reviewed scientific community – Szulczewski et al. (2012) and Zhou et al. (2008). A statistical analysis of the estimates generated by multiple methods revealed that assessments of CO_2 storage potential made at the prospective level were often statistically indistinguishable from each other, implying that the differences in methodologies are small with respect to the uncertainties in the geologic properties of storage rock in the absence of detailed site-specific characterization.

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1. Introduction

Estimates of CO_2 storage potential in geologic formations at the national, regional, basin, and formation scale provide critical information to base broad energy-related government policy and business decisions for investment in Carbon Capture, Utilization, and Storage (CCUS) technologies to mitigate CO_2 emissions. These estimates also aid in identifying potential regions to successfully implement CCUS technologies. No standard method has been adopted to estimate storage potential and geologic data for saline formations are sparse. Several methods have been proposed to estimate CO_2 storage potential. In addition, terminology and scale describing one estimate is often inconsistent when comparing with other estimates, adding more uncertainty. As characterization of specific sites develops and detailed geologic data is obtained, these preliminary estimates will be refined and uncertainty will be reduced. However, until detailed geologic characterization is available for CO_2 storage estimates to advance to the site-specific scale, dependable CO_2 storage estimates at the prospective scale are essential to properly advise decision makers and facilitate widespread deployment of CCUS technologies.

1.1. Saline formations

The DOE has identified saline formations as layers of porous rock that are saturated with brine (US-DOE-NETL, 2012). They have enormous storage potential and are more extensive than oil and gas fields and coals seams. However, because these formations do not produce hydrocarbons, less is known about their geologic properties. When estimating CO_2 storage potential in saline formations, uncertainty is high because simplifying assumptions are often made because of a deficiency or absence of geologic data and an inability to account for the natural heterogeneity.

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1.2. Development of CO_2 storage methods

Early storage estimates compiled by the Intergovernmental Panel on Climate Change in 2005 are associated with a high degree of uncertainty (IPCC, 2005). The upper limit for deep saline formations, for example, was reported as "uncertain, but could possibly be 1×10^4 GT of CO_2". As discussed by Bradshaw et al. (2007), early storage estimates applied by various researchers used inconsistent methods that did not allow for comparison. These estimates were highly variable and often contradictory. In some cases worldwide estimates were smaller than country estimates. Recently, initiatives for assessing CO_2 geologic storage potential that have been conducted since 1993 were summarized (Dooley, 2011).

To address inconsistent storage estimates, the Carbon Sequestration Leadership Forum (CSLF) created a task force in 2005 to develop a consistent method for estimating CO_2 storage in deep saline aquifers, coal beds, and hydrocarbon reservoirs. This method was formally published in CSLF reports (CSLF, 2010) and a peer-reviewed article (Bachu et al., 2007). The boundary conditions were assumed to be open where the native formation fluids would be displaced upon injection of CO_2 and pressure build-up would be dissipated. This method was widely accepted. Most of today's world storage estimates, with the exception of the United States, are based on the CSLF method. Some recent examples of by application of the CSLF method worldwide include the following: "Estimation of CO_2 aquifer storage potential in Japan" (Takahashi et al., 2009), "Assessment of potential for geological storage of carbon dioxide in Ireland and Northern Ireland" (Lewis et al., 2009), and "CO_2 storage capacity of deep aquifers and hydrocarbon fields in Poland–EU Geo Capacity project results" (Radoslaw et al., 2009). In addition, the North American Carbon Atlas Partnership (NACSA, 2012) which includes Canada, the U.S., and Mexico in part applies the CSLF method for CO_2 storage estimates.

The United States Department of Energy's (US-DOE) Carbon Storage program also developed a CO_2 storage method in conjunction with its Regional Carbon Sequestration Partnerships (RCSPs) for volumetric CO_2 storage in oil and gas reservoir, deep saline formations, and unmineable coal seams in the United States and portions of Canada (US-DOE-NETL, 2012). The boundary conditions were also considered to be open as with the CSLF method. The US-DOE's National Energy Technology Laboratory (NETL) published CO_2 storage estimates in the 2007 and 2008 Carbon Sequestration Atlas of the United States and Canada (US-DOE-NETL, 2007, 2008). This methodology was updated and underwent extensive external review. Refined CO_2 storage estimates were published in the 2010 and 2012 Carbon Sequestration Atlas of the United States and Canada (US-DOE-NETL, 2010a, 2012). While the CSLF and US-DOE methods were developed independently, the CSLF Task Force (CSLF, 2008) and IEA-GHG (IEA-GHG, 2009) reported the CSLF and US-DOE methodologies to be equivalent, with the exception that the CSLF methodology accounts directly for residual gas saturation while the US-DOE method accounts for residual gas saturation stochastically.

In 2007, the United States Geologic Survey as authorized by the Energy Independence and Security Act (Public Law 110-140) began developing a CO_2 storage method to conduct a national assessment of the United States in conjunction with the Environmental Protection Agency (EPA) and DOE. As with the CSLF and US-DOE methods, the USGS also assumed the boundary conditions would be open. They released their first report describing the assessment method in 2010 (Brennan et al., 2010) and a second report with method updates and details of the method implementation in 2013 (Blondes et al., 2013). The CO_2 storage assessment results for the conterminous United States were recently released in June of 2013.

CO_2 Geological Storage Solutions in Australia also developed a method for CO_2 storage based on migration assisted storage. They consider saline formations, depleted oil and gas fields, and deep unmineable coal seams. This method was recently compared to the US-DOE and USGS methods (Popova et al., 2012). Storage estimates for Australia can be found in the 2009 Queensland CO_2 Geological Storage Atlas (Bradshaw et al., 2011).

Other methods were developed for CO_2 storage that consider closed or semi-closed boundaries in which saline formations are surrounded by low permeability seals were native fluids cannot be displaced upon CO_2 injection (Economides and Ehlig-Economides, 2009; van der Meer and Yavuz, 2009; Zhou et al., 2008). In these cases, over-pressurization will need to be addressed upon injection of CO_2 to prevent damage to the formation. Zhou et al. (2008) developed a detailed quick assessment method for CO_2 storage in closed and semi-closed systems where they account for the compressibility of the formation rock and fluids and the maximum pressure build-up to avoid geo-mechanical damage to the formation. Recently, Szulczewski et al. (2012) released a storage method that accounts for fluid dynamics and injection rates and provides CO_2 storage estimates for saline formation throughout the United States. Because they examine both CO_2 migration and pressure build-up, this method is applicable to formations with open or closed boundaries.

1.3. CO_2 storage terminology and classification

Currently there is no standard classification procedure to categorize CO_2 storage estimates developed by a particular method. The CSLF has proposed a Techno-Economic Resource Pyramid where CO_2 storage can be categorized into theoretical, effective, practical and matched capacity (Bachu et al., 2007). The US-DOE applies a CO_2 Storage Classification Framework analogous to the exploration and development of oil and gas accumulations. In this case, CO_2 storage can be categorized as prospective storage resource, contingent storage resource, or capacity (US-DOE-NETL, 2010b). CO_2 resource is defined as the pore volume within the characterized geologic formation and capacity is defined as the quantity of CO_2 anticipated to be commercially storable by available technology applied to known formations from a given date forward under defined conditions and includes economic, regulatory, and legal constraints. The IEA-GHG advised a storage classification with CO_2 categories common to both the CSLF and US-DOE (IEA-GHG, 2009). IEA-GHG categories include Theoretical Storage Resource, Characterized Storage Resource, Effective Storage Resource, Practical Storage Capacity, Contingent Storage Resource, Unusable Storage Resource, and Uncharacterized Storage Resource. For the methods described above, the CSLF reports effective storage capacity, the US-DOE reports prospective storage resource, the USGS reports technically accessible storage resource, the quick assessment method reports CO_2 storage capacity (Zhou et al., 2008), and Szulczewski et al. (2012) report CO_2 migration-limited and pressure-limited capacity.

1.4. Storage efficiency and mechanism

Efficiency of storage of CO_2 in saline formations is a ubiquitous term found in all storage methods. Typically this term defines a fraction of the pore space that will be occupied by the injected CO_2 (Goodman et al., 2011). In general, a generic efficiency value can be broadly applied to formations of interest. In 1992, it was assumed that 2% of the pore volume could be used as storage space, however, this value is arbitrary and may lead to unreliable results (van der Meer and Yavuz, 2009). A specific efficiency can be calculated for individual formations based on site and geologic parameters. Efficiency is dependent on boundary conditions of the formation. In open systems, the fraction of accessible pore volume is typically estimated by the use of displacement terms (Lake, 1989). In closed systems, the fraction of accessible pore volume is

estimated by compressibility and pressure constraints (Zhou et al., 2008). Efficiency can also be approximated with capillary trapping numbers and mobility factors (Szulczewski and Juanes, 2009). The efficiency term also accommodates CO_2 trapping mechanisms such as structural and stratigraphic, hydrodynamic, residual, dissolution, mineralization and adsorption trapping; lithology properties such as carbonates, clastics, and limestone; and rock permeability classes.

As for the methods described above, each method accounts for specific CO_2 storage mechanisms and applies an efficiency that's unique to the method. The CSLF method considers structural and stratigraphic trapping. Efficiency values, while suggested, were not proposed at the time of publication. The US-DOE method considers structural and hydrodynamic trapping and applies efficiency based on lithology. The USGS considers buoyant and residual trapping and applies efficiency as a function of permeability class. Their updated method applies efficiency as a function of depth for residual trapping (Blondes et al., 2013). Szulczewski et al. (2012) consider residual and solubility trapping and calculate efficiency specific to each formation based on its geologic properties. For the quick assessment method by Zhou et al. (2008), efficiency is based on reservoir compressibility properties and sustainable pressure of the formation.

In summary, there are multiple methods that have been applied to estimate CO_2 storage in saline formations. Each method applies its own terminology, mechanisms, efficiency terms, and sometimes its own classification scheme. Table 1 compiles key information for several of the most high-profile CO_2 storage assessment methods to aid in clarifying the approach, year developed, boundary condition applied, terminology, and how efficiency and storage mechanisms are addressed. While dependable CO_2 storage estimates are essential to policy and business decisions, an increasing number of storage estimates calculated by different and potentially conflicting methods makes it is difficult to assess the uncertainty of these estimates without knowing how the methods compare to one another.

In this study, we compare methods to estimate CO_2 storage in saline formations to determine if the method used for estimation of storage resource significantly impacts the results. This study compares estimates produced by methods using the same input data to assess significant differences in estimates generated by the various methods. The following items are addressed in sequence in this study:

(1) The following six methods available to the public were selected (Section 2).
   (i) CSLF: Bachu et al. (2007): CO_2 storage capacity estimation: methodology and gaps.
   (ii) US-DOE (2007, 2008): U.S. DOE methodology for the development of geologic storage estimates for carbon dioxide.
   (iii) US-DOE (2010, 2012): U.S. DOE methodology for the development of geologic storage estimates for carbon dioxide.
   (iv) USGS: Brennan et al. (2010): a probabilistic assessment methodology for the evaluation of geologic carbon dioxide storage.
   (v) Szulczewski et al. (2012): lifetime of carbon capture and storage as a climate-change mitigation technology.
   (vi) Zhou et al. (2008): a method for quick assessment of CO_2 storage capacity in closed and semi-closed saline formations.
(2) A common saline data set that includes thirteen formations was adapted to provide the geologic inputs required by each assessment method (Section 3).
(3) The six methods were applied using the same geologic input parameters to thirteen saline formations to estimate CO_2 storage potential (Section 4)
(4) CO_2 storage estimates were compared mathematically to determine if any of the methods produce statistically significant differences (Section 5).
(5) Implications of CO_2 storage method choice were discussed (Section 6).

2. Description of CO_2 storage methods

Six CO_2 storage methods were selected for this comparison study. The methods applied include those by (1) international efforts – the Carbon Sequestration Leadership Forum (Bachu et al., 2007); (2) United States government agencies – U.S. Department of Energy-National Energy Technology Laboratory (US-DOE-NETL, 2012) and United States Geological Survey (Brennan et al., 2010); and (3) the peer-reviewed scientific community – Szulczewski et al. (2012) and Zhou et al. (2008). These methods are available to the public, referenced in the literature, and transparent in that all input parameters are defined and could be readily applied to an existing, uniform data set. All methods target estimates of CO_2 storage resource at a high-level such as the national, regional, basin or formation scale. Site-specific CO_2 storage methods are not considered in this study. Economic, legal, and regulatory considerations are not included in these methods. Injectivity issues are considered by one method (Szulczewski et al., 2012). All methods assume that the depth is sufficient to store CO_2 as a supercritical fluid, a seal system is present to limit vertical movement of CO_2 to the surface, and that hydro-geologic conditions will adequately contain the CO_2 within the formation. In this study, methods that apply both open and closed boundary conditions were considered.

A brief description of the methods applied in this study is supplied below:

2.1. CO_2 storage capacity estimation: methodology and gaps (CSLF: Bachu et al., 2007)

This method stems from the Technical Group Taskforce on CO_2 storage estimates led by the Carbon Sequestration Leadership Forum (CSLF) (Bachu et al., 2007). The CSLF defines itself as "a ministerial-level international climate change initiative that is focused on the development of improved cost-effective technologies for the separation and capture of carbon dioxide (CO_2) for its transport and long-term safe storage" (CSLF, 2010). The CSLF effort targets types of geological media, operating time frames, resource-reserve pyramid classification, and assessment of scale and resolution.

For saline aquifers, the boundary conditions are considered to be open. The general equation used for storing CO_2 in structural and stratigraphic traps as static trapping is:

$$M_{CO_2e} = V_{trap}\phi(1 - S_{wirr})\rho CO_2 C_c = Ah\phi(1 - S_{wirr})\rho CO_2 C_c \quad (1)$$

and defined by Bachu et al. (2007) as "M_{CO2e} is the effective storage capacity as designated in the resource-reserve classification pyramid by CO_2 mass representing a subset of the theoretical capacity. This estimate usually changes with the acquisition of new data and or knowledge. V_{trap} is the geometric volume of the structural or stratigraphic trap down to the spill point, ϕ is the porosity, S_{wirr} is the irreducible water saturation, ρCO_2 is the CO_2 density at the temperature and pressure of the aquifer, C_c is a capacity coefficient that incorporates the cumulative effects of trap heterogeneity, CO_2 buoyancy and sweep efficiency, and A and h are the trap area and average gross thickness, respectively." At the time of publication of Bachu et al. (2007), no values in the literature were available for this capacity coefficient as the CSLF deems capacity coefficients as "site-specific and must be determined through numerical simulations and/or field work". Recent work by the International Energy Agency Greenhouse Gas (IEA-GHG) R&D Program cites C_c at the

Table 1
Overview of CO_2 storage methods.

Method	Year	Volumetric approach	Boundary	Terminology	Trapping mechanism	Efficiency
CSLF	2007	√	Open	Effective capacity	Structural and stratigraphic	Field[a]
US-DOE	2007, 2008	√	Open	Prospective storage resource	Structural and hydrodynamic	Generic[b]
US-DOE	2010, 2012	√	Open	Prospective storage resource	Structural and hydrodynamic	Lithology[c]
USGS	2010, 2013	√	Open	Technically assessable storage resource	Buoyant and residual	Permeability[d]
Szulczewski et al.	2012	√	Open and closed	Migration-limited and pressure-limited capacity	Residual and solubility	Formation Specific[e]
Zhou et al.	2008	√	Closed	Storage capacity	Compressibility	Compressibility[f]

[a] To be determined through field work or numerical simulation.
[b] Uniform value for all formations.
[c] Based on formation lithology.
[d] Based on rock permeability class.
[e] Based on geologic properties of formation.
[f] Based on formation pressure and compressibility constraints.

formation and site-specific scale specific to this method. Formation (C_c) values range between 2.4% and 10% based on their 10th to 90th percentiles (IEA-GHG, 2009).

2.2. U.S. DOE methodology for the development of geologic storage potential for carbon dioxide at the national and regional scale US-DOE (2007, 2008) and US-DOE (2010, 2012)

The U.S. Department of Energy (US-DOE) method as detailed in Goodman et al. (2011) "is intended for external users such as the Regional Carbon Sequestration Partnerships (RCSP), future project developers, and governmental entities to produce prospective CO_2 resource assessments of potential CO_2 storage reservoirs in the United States and Canada at the regional and national scale (Goodman et al., 2011; US-DOE-NETL, 2007, 2008, 2010a, 2012). The method, however, is general enough to be applied globally. Structural and hydrodynamic trapping are initially the dominant trapping mechanisms and are the focus of the US-DOE method." The CO_2 storage estimates are categorized to be prospective storage resources according to the DOE storage classification structure (US-DOE-NETL, 2010b). The US-DOE (2007, 2008) method was developed for the first and second editions of the Carbon Sequestration Atlas of the United States and Canada; it assumes open-boundary conditions (US-DOE-NETL, 2007, 2008). The US-DOE (2010, 2012) method was developed for the third and fourth editions of the Carbon Sequestration Atlas; it assumes open-boundary conditions (US-DOE-NETL, 2010a, 2012).

The volumetric equation to calculate the CO_2 storage resource mass estimate (G_{CO_2}) for geologic storage in saline formations for US-DOE (2007, 2008) and US-DOE (2010, 2012) is:

$$G_{CO_2} = A_t h_g \phi_{tot} \rho E_{saline} \quad (2)$$

where as detailed in Goodman et al. (2011), "A_t is the total geographical area that defines the basin or region being assessed for CO_2 storage, h_g is the gross thickness of the saline formation for which CO_2 storage is assessed within the basin or region defined by A_t, ϕ_{tot} is the total porosity in volume defined by the net thickness, ρ is the density of CO_2 evaluated at the pressure and temperature that represents storage conditions anticipated for a specific geologic unit averaged over h_g and A_t."

According to Goodman et al. (2011), "carbon dioxide storage efficiency [E_{saline}] gauges the fraction of the accessible pore volume that will be occupied by the injected CO_2." In the US-DOE (2007, 2008) method, efficiency is averaged over four stochastic simulation cases that are based on (1) normal distributions with one standard deviation, (2) normal distributions with one half standard deviation, (3) uniform distributions, and (4) a mixture of distributions such as normal, lognormal, and uniform. For the US-DOE (2007, 2008) method, efficiency ranged between 0.7% and 4.7% based on their 15th to 85th percentiles (US-DOE-NETL, 2007, 2008). In the US-DOE (2010, 2012) method, efficiency was estimated by the Log Odds method with Monte Carlo sampling and ranged between 0.40% and 5.5% based on the 10th to 90th percentiles as a function of lithology (US-DOE-NETL, 2010a, 2012). Both the CLSF (2008) and IEA-GHG (2009) determined that the CSLF and US-DOE methods produce similar results as long as the same assumptions are applied to each method. Worldwide CO_2 storage estimates are primarily based on either CSLF or US-DOE methods.

2.3. A probabilistic assessment methodology for the evaluation of geologic carbon dioxide storage (USGS: Brennan et al., 2010)

As discussed by Brennan et al. (2010), "In 2007, the Energy Independence and Security Act (Public Law 110-140) authorized the United States Geological Survey (USGS) to conduct a national assessment of potential geologic storage resources for CO_2 in cooperation with the U.S. Environmental Protection Agency (EPA) and the DOE." As a result of this legislation, USGS developed a method that is intended to be used by the USGS's geologists for assessments of CO_2 storage resource in the United States "at scales ranging from regional to sub-basinal in which storage assessment units are defined on the basis of common geologic and hydrologic characteristics (Brennan et al., 2010)." The USGS updated USGS method and released CO_2 storage assessments stemming from this Act in June of 2013 (Blondes et al., 2013). The updated 2013 USGS method was not included in this work as this study had already undergone the peer-review process when the updated method and estimates were released by the USGS.

As described by Brennan et al. (2010), this method estimates "the technically accessible storage resource, which is defined as the mass of CO_2 that can be stored in the pore volume of the saline storage formation." The boundary conditions are considered to be open. Brennan et al. (2010) also states that "this resource is the product of CO_2 density and storage volume, which is the fraction of the pore volume that will retain injected CO_2." The storage formation pore volume (SF_{PV}) is calculated for the pore volume available for buoyantly trapped CO_2 and residually trapped CO_2.

$$SF_{PV} = A_{SF} T_{PI} \phi_{PI} \quad (3)$$

where according to Brennan et al. (2010), "A_{SF} is the mean area of storage assessment, T_{PI} is the mean thickness of the porous interval, and ϕ_{PI} is the mean porosity." The buoyant trapping and residual trapping are calculated separately because they are modified by different geologic uncertainties and storage efficiencies. Brennan et al. (2010) describes "storage efficiency as the value that represents the fraction of the total available pore space that will retain free-phase CO_2." Buoyant trapping efficiencies range between 10% and 60% and residual trapping efficiencies range between 1% and 15% by rock class based on their 5th and 95th percentiles. Brennan et al. (2010) states that, "the technically accessible storage resource is determined by multiplying the TA_{SV} [technical accessible storage volume] by the density of CO_2."

Lifetime of carbon capture and storage as a climate-change mitigation technology (Szulczewski et al., 2012).

According to Szulczewski et al. (2012), "the storage capacity of a saline aquifer is the maximum amount of CO_2 that could be injected and securely stored under geologic constraints, such as the aquifer's size and the integrity of its caprock. Both residual trapping, in which zones of CO_2 become immobilized by capillary forces, and solubility trapping, in which CO_2 dissolves into the groundwater are considered." This method is applicable to both open and closed systems depending upon the geology of the formation of interest. Szulczewski et al. (2012) as describes "how much CO_2 can be trapped in the pore space of the formation is partitioned by CO_2 trapping and migration-limited capacity, a time independent component, and CO_2 capacity due to limitations on the injection rate through pressure dissipation and pressure-limited capacity, a time dependent component." Thus, the total CO_2 storage supply estimate is determined by the more conservative capacity curve: the dissipation and pressure-limited curve or the CO_2 trapping and migration-limited curve. This method reports CO_2 storage capacities at several injection times between 25 years and 600 years.

The mass of CO_2 (C_t) stored for the CO_2 trapping and migration-limited method at the basin scale, is calculated by

$$C_t = \rho_g L_t W H \phi (1 - S_{wc}) \frac{2}{\varepsilon_T} \tag{4}$$

where according to Szulczewski et al. (2012), "ρ_g is the CO_2 density, L_t is the length of domain for migration model, W is the width of the well array, H is the net aquifer thickness, ϕ is the porosity, S_{wc} is the connate water saturation, and $2/\varepsilon_T$ is the efficiency factor. The efficiency factor is calculated using a reservoir model to simulate CO_2 migration through an aquifer as well as solubility and capillary trapping." The mass of CO_2 (C_p) stored for the CO_2 dissipation and pressure-limited method at the basin scale, is calculated by:

$$C_p = \rho_g H W \sqrt{\frac{k_{ap} c T}{\mu_w}} \frac{P_{fraq} - (\rho_o + \overline{\rho_w} g D)}{4 \tilde{p}_{max}} \tag{5}$$

"where ρ_g is the CO_2 density, H is the net aquifer thickness, W is the width of the well array, k_{aq} is the aquifer permeability, c is the compressibility, T is the temperature, μ_w is the brine viscosity, P_{fraq} is the fracture pressure, ρ_o is the density, ρ_w is the brine density, g is the gravitational acceleration, D is the depth to the top of aquifer, and p_{max} is maximum dimensionless pressure."

2.4. A method for quick assessment of CO_2 storage capacity in closed and semi-closed saline formations (Zhou et al., 2008)

As described in detail by Zhou et al. (2008), this is a simple method for assessing the storage capacity of closed and semi-closed storage formations. In this comparison study, only closed systems were considered. Zhou et al. (2008) describes "the basic principle is that CO_2 injection into these systems will lead to pressure build-up, because an additional volume of fluid needs to be stored. The injected CO_2 displaces an equivalent volume of native brine, which may either be stored in the expanded pore space due to compression of the rock, or be stored in the expanded pore space in the seals. The method is designed to provide capacity estimates at early stages of site selection and characterization, when quick assessments of multiple sites may be needed and site characterization data is sparse." According to Zhou et al. (2008), "In a closed system for saline formations, the available volume for storage of CO_2 is provided only by the expansion of the pore volume in the formation and seals and the increased brine density in response to pressure buildup in the storage formation." The storage capacity by CO_2 mass for the closed systems ($M_{CO_2}(t_I)$) is:

$$M_{CO_2}(t_1) = (\beta_p + \beta_w) \Delta P(t_1) \rho V_f = (\beta_p + \beta_w) \Delta P(t_1) \rho \phi_f A B_f \tag{6}$$

where β_p and β_w are the pore compressibility and native brine compressibility, respectively, ΔP is the sustainable pressure build-up at time t_1, and ρ is the CO_2 density at temperature and pressure of the formation. V_f is the volume of the formation and is equal to the initial porosity (ϕ_f), horizontal area (A), and gross thickness (B_f) of the formation. The storage efficiency (E) in a closed system is derived by simplifying Eq. (7).

$$E = (\beta_p + \beta_w) \Delta P(t_1) \tag{7}$$

3. Description of general saline formation data sets

A general saline data set that includes thirteen formations labeled A through M was created to provide the geologic inputs needed by each assessment method. The saline formation data sets were based on the saline formations detailed by Szulczewski et al. (2012) which included formations in the United States where the data is publicly available. It is important to note that data for these saline formations were sparse due to a deficiency of detailed geologic characterization information, requiring wide uncertainty bands to account for the natural heterogeneity. As described in Szulczewski et al. (2012), they "selected rectangular volumes representative of various formations found in the United States that met the following criteria: (1) the depth must exceed 800 m so that CO_2 is stored efficiently as a high-density, supercritical fluid; (2) the aquifer and caprock must be laterally continuous over long distances; and (3) there must be very few faults that could serve as leakage pathways." In this work, we adapted saline data sets to provide geologic parameters needed by the different storage methods. Formations with water that contained total dissolved solids less than 1% (10,000 ppm) were omitted in order to concur with drinking water regulations stated by the EPA (2010). Location names of the saline formations were not used in order to focus the study on comparing CO_2 storage resource estimates and to avoid implying endorsement to specific saline formations.

Key saline formation geologic parameters are compiled in Table 2 for general saline formation data sets A through M. Length, width, net thickness, porosity, CO_2 density, and permeability values were taken directly from the saline formations detailed by Szulczewski et al. (2012). Detailed data and formation properties regarding these parameters can be found in Szulczewski et al. (2012) and in the Supplementary Section. The methods by CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), and Zhou et al. (2008) require gross thickness, while the methods by USGS: Brennan et al. (2010) and Szulczewski et al. (2012) require net thickness. Because gross thickness was not compiled by Szulczewski et al. (2012), we needed to provide an estimate. Gross thickness is the average thickness of the saline formation for storage of CO_2 in a defined area. Net thickness is a portion of the gross thickness that meets a set of requirements for the practical storage of CO_2 based on porosity and permeability. For methods that use gross thickness, a net-to-gross thickness efficiency factor is used to discount the gross thickness to account for these practical

Table 2
Key saline formation parameters.

Parameter	Symbol	Unit	Formation												
			A	B	C	D	E	F	G	H	I	J	K	L	M
Length[a]	L_t	km	200	100	50	300	40	50	90	90	30	60	50	70	100
Width[a]	W	km	200	200	100	200	100	100	200	100	200	100	100	80	100
Net thickness[a]	H	m	200	400	2000	100	900	700	500	200	200	600	200	15	40
Gross thickness[b]	H_g	m	416	831	4156	208	1870	1455	1412	416	416	1694	416	31	83
Porosity[a]	ϕ		0.2	0.2	0.2	0.2	0.2	0.2	0.08	0.2	0.2	0.08	0.2	0.2	0.06
CO_2 density[a]	ρ	kg/m^3	800	700	500	700	500	500	500	700	600	500	600	600	800
Permeability[a]	k	mD	100	100	400	100	400	400	60	900	100	60	100	300	50
Pore compressibility[c]	B_p	GPa^{-1}	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Brine compressibility[d]	B_w	GPa^{-1}	0.35	0.37	0.51	0.33	0.44	0.38	0.51	0.38	0.48	0.47	0.49	0.39	0.36
Pressure buildup[e]	ΔP	Mpa	10	10	5	5	5	5	10	5	15	15	15	10	10

[a] Baseline values from Szulczewski et al. (2012).
[b] Calculated by dividing the baseline net thickness by the mean net-to gross thickness efficiency term [0.48 for clastics, 0.41 for dolomite, and 0.35 for limestone].
[c] Set to range between 1×10^{-10} Pa^{-1} and 5×10^{-10} Pa^{-1}.
[d] Calculated as described by Battistelli et al. (1997).
[e] 50% of the initial hydrostatic pressure at the top of the storage formation.

requirements. As described in the US-DOE method, "net-to-gross thickness efficiency is estimated by Monte Carlo sampling for saline formations based directly on the 10th and 90th percentiles of net-to-gross thickness as reported by *IEA-GHG* (Goodman et al., 2011)." By applying the US-DOE method, we calculated a mean net-to-gross thickness efficiency for clastics (0.48), dolomite (0.41), and limestone (0.35) lithologies. We then approximated gross thickness by dividing net thickness provided by Szulczewski et al. (2012) by the mean net-to gross thickness efficiency term based on lithology reported above.

Pore compressibility, brine compressibility, and sustainable pressure build-up are required by the closed system method authored by Zhou et al. (2008). Pore compressibility varies widely depending upon subsurface material. Szulczewski et al. (2012) set an average compressibility for each formation to 1×10^{-10} Pa^{-1} because little or no data was available for compressibility of the aquifers and caprocks chosen in their work. The IEA-GHG (2009) reported pore compressibility between 2.8×10^{-10} Pa^{-1} and 6.9×10^{-10} Pa^{-1}. Zhou et al. (2008) reported pore compressibility for deep formations ranging from (4.5×10^{-11} to 4.5×10^{-9}) Pa^{-1}. Others reported compressibility for sandstones and limestones between (1×10^{-11} and 1×10^{-10}) Pa^{-1} (Domenico and Schwartz, 1998; Fjar et al., 1991). In this study, low and high values or pore compressibility were set at 1×10^{-10} Pa^{-1} and 5×10^{-10} Pa^{-1}, respectively. Brine compressibility for each formation was directly calculated while assuming the brine could be represented by a sodium chloride solution (Battistelli et al., 1997). Sustainable pressure build-up refers to a pressure that the storage formation can accommodate without undergoing geo-mechanical damage (Neuzil, 2003). By following the example by Zhou et al. (2008), sustainable pressure build-up was calculated by taking "50% of the initial hydrostatic pressure at the top of the storage formation." The initial hydrostatic pressure was calculated by multiplying the depth at the top of the formation by the hydrostatic pressure gradient for fresh water (0.009792 MPa/m) (Schlumberger, 2012). All geologic parameters described above are compiled in Table 2 and the Supplementary Section.

4. Application of CO_2 storage methods to general saline formation data sets

The authors worked with the corresponding authors of the methods applied in this study to ensure that each method was applied to the general saline formation data sets correctly.

4.1. CSLF: Bachu et al. (2007) method

CO_2 storage estimates from the CSLF: Bachu et al. (2007) method were calculated from Eq. (1) by applying the method deterministically for saline formation datasets A through M summarized in Table 2. Baseline length (L_t) multiplied by width (W) from Table 2 produced trap area (A), gross thickness (H_g) from Table 2 was used as average gross thickness (h), porosity (ϕ) and CO_2 density (ρ) were used directly from Table 2, and the S_{wirr} is the irreducible water saturation was set to 0.4 for all formation data sets based on data supplied by Szulczewski et al. (2012). Capacity coefficients, $C_c^*(1 - S_{wirr})$, were used directly from the IEA-GHG (2009) as a function of lithology based on their 10th, 50th, and 90th percentiles: Clastics (1.86, 2.70, and 6.00)%; Dolomite (2.58, 3.26, and 5.54)%; and Limestone (1.41, 2.04, and 3.27)%.

4.2. US-DOE (2007, 2008) and US-DOE (2010, 2012) method

CO_2 storage resource estimates from the US-DOE (2007, 2008) and US-DOE (2010, 2012) methods were calculated by Monte Carlo simulation using the program GoldSim (see Supplementary Section for description of the GoldSim Model). Baseline values from Table 2 for parameters A_t, h_g, ϕ_{tot}, and ρ shown in Eq. (2) were applied to the US-DOE (2007, 2008) and US-DOE (2010, 2012) methods for each saline formation data set. For US-DOE (2007, 2008), E_{saline} values applied were (0.7, 2.1, and 4.7)% based on their 15th, 50th, and 85th percentiles. For US-DOE (2010, 2012), E_{saline} values applied were based on lithology at their 10th, 50th, and 90th percentiles and are as follows: Clastics (0.5, 2.0 and 5.4)%; Dolomite (0.64, 2.2, and 5.5)%; and Limestone (0.40, 1.5, and 4.1)%.

4.3. USGS: Brennan et al. (2010) method

For the USGS: Brennan et al. (2010) method, all parameters were treated stochastically in the Monte Carlo simulation program GoldSim (See Supplementary Section for GoldSim Model description). Minimum, baseline, and maximum values in a triangular distribution were applied to A_{SF}, T_{PI}, ϕ_{PI}, CO_2 density, and residual trapping efficiency according to Eq. (3). Table 2 provides baseline values for saline formations A through M. Minimum and maximum values were calculated by the uncertainty analysis provided by Szulczewski et al. (2012) and are tabulated in the Supplementary Section. Buoyant trapping was set to 0% because the caprock of the selected rectangular volumes of the saline formation data sets created by Szulczewski et al. (2012) were assumed to be

linear ensuring that structural trapping would not occur at the top of an anticline or in a tilted fault block. Minimum, baseline, and maximum values for residual trapping efficiency as a function of permeability are as follows: Class 1 with permeability greater than 0.9869233 μm^2 (1 d): (1, 5, and 7)% efficiency, Class 2 with permeability between 0.9869233×10^{-3} μm^2 and 0.9869233 μm^2 (1 md and 1000 md): (1, 7, and 15)% efficiency, and Class 3 with permeability less than 0.9869233×10^{-3} μm^2 (1 md): (0, 0, and 7)% efficiency. Eleven saline data formation sets were found to have Class 2 permeability. Two formations were found to have Class 1 and 2 permeability ranges where it was assumed that Class 1 accounted for 5% and Class 2 accounted for 95% of the permeability range. Technically accessible CO_2 storage estimates are reported at their 5th, 50th, and 95th percentiles.

4.4. Szulczewski et al. (2012) method

This comparison effort is focused on time-independent storage resource in the absence of injectivity limitations. Therefore, only the CO_2 trapping and migration-limited capacities provided by Szulczewski et al. (2012) outlined by Eq. (4) were considered. This is equivalent to the reported capacities by Szulczewski et al. (2012) in the limit of long injection times. The storage volumes reported by Szulczewski et al. (2012) had reached the volumetric limit by 300 years for 12 of the 13 formations used in this comparison. For the 13th, the time-independent resource was communicated to us directly by the authors. In this comparison effort, the CO_2 trapping and migration-limited CO_2 storage resource estimates were used directly from the article by Szulczewski et al. (2012) for each saline formation considered. Szulczewski et al. (2012) reported baseline storage capacities based on a normal distribution where one standard deviation is used as an appropriate measure of uncertainty from the probability density function. Because of the assumed normality of the distribution, one standard deviation was taken to be the half-range of the 16th and 84th percentiles of baseline storage capacity.

4.5. Zhou et al. (2008) method

Zhou et al. (2008) storage resource estimates were calculated stochastically by the Monte Carlo simulation program GoldSim (See Supplementary Section for GoldSim Model description). The parameters, β_p, β_w, ρ ϕ_f, A, and B_f, were treated stochastically assuming the values for each quantity follow a triangular distribution and ΔP was treated deterministically according to Eq. (6) (see Table 2 for baseline values for saline formations A–M). CO_2 storage estimates are reported at their 10th, 50th, and 90th percentiles. Baseline length (L_t) multiplied by width (W) from Table 2 produced horizontal (A), gross thickness (H_g) from Table 2 was used as gross thickness (B_f), porosity (ϕ) and CO_2 density (ρ) were used directly from Table 2. Minimum and maximum values were calculated by the uncertainty analysis provided by Szulczewski et al. (2012) and are tabulated in the Supplementary Section. Pore compressibility (B_p) was set to range between 1×10^{-10} Pa^{-1} and 5×10^{-10} Pa^{-1}, brine compressibility (B_w) for each formation data set was directly calculated as described by Battistelli et al. (1997), and sustainable pressure build-up (ΔP) was calculated by following the example by Zhou et al. (2008) where "50% of the initial hydrostatic pressure at the top of the storage formation" is equal to the sustainable pressure.

5. Comparison CO_2 storage methods applied to general saline formation data sets

5.1. Comparison of CO_2 storage estimates by multiple methods

This study applied multiple CO_2 storage methods to a uniform saline data set consisting of thirteen saline formations to understand how method choice impacts CO_2 storage estimates. Saline formation inputs for length, width, porosity, and CO_2 density were consistently applied to each method. Gross or net thickness was applied as prescribed by the method. As discussed in detail in Section 5.3, the efficiency term was different for each method and uniquely applied.

CO_2 storage estimates calculated by applying the methods by CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), USGS: Brennan et al. (2010), Szulczewski et al. (2012), and Zhou et al. (2008) to thirteen saline formation data sets labeled A–M are shown in the clustered column graphs in Fig. 1a and b. The individual vertical columns represent the middle CO_2 storage estimates, which are based on their 50th percentile. The black solid error bars coinciding with each column represent low and high estimates of CO_2 storage at the 10th and 90th percentiles, respectively. Values for middle CO_2 storage estimates, which are based on their 50th percentile, for formations A through H are tabulated in Table 3. Alternative CO_2 storage percentiles are compiled in the Supplementary Section.

The CO_2 storage resource estimates at the 50th percentile were first compared qualitatively. The closed-boundary method by Zhou et al. (2008), typically, reports the lowest CO_2 storage estimate and the open-boundary method by USGS: Brennan et al. (2010), typically, reports the highest CO_2 storage estimate. The open-boundary estimates by CSLF: Bachu et al. (2007), US-DOE (2007, 2008), and US-DOE (2010, 2012) fall between the open-boundary estimates by USGS: Brennan et al. (2010) and closed-boundary estimates by Zhou et al. (2008) for CO_2 storage resource. It is important to note

Table 3
Prospective CO_2 storage resource estimates (GT)[a].

Formation	CSLF: Bachu et al. (2007) open	US-DOE (2007, 2008) open	US-DOE (2010, 2012) open	USGS: Brennan et al. (2010) open	Szulczewski et al. (2012) migration-limited	Zhou et al. (2008) closed
A	72	55	52	95	10	17
B	63	48	46	84	88	15
C	56	43	41	75	18	9
D	47	36	34	63	17	6
E	20	15	15	27	9	3
F	20	15	14	26	12	2
G	21	21	15	27	7	8
H	14	11	10	19	14	2
I	8	6	6	11	5	4
J	8	8	6	11	5	5
K	7	5	5	9	4	3
L	1	0.4	0.4	1	2	0.1
M	1	1	1	1	2	0.3

[a] CO_2 storage resource estimates based on 50th percentile.

that if the same assumptions were applied to the CSLF and US-DOE methods, the CO_2 storage estimates would be equivalent (CSLF, 2008; IEA-GHG, 2009). In most cases, migration limited estimates by Szulczewski et al. (2012) are similar to the closed-boundary estimates provided by Zhou et al. (2008) for CO_2 storage.

The CO_2 storage estimates were quantitatively compared with mathematical rigor to determine if any of the methods produce statistically significant differences. In these analyses, a statistically significant result is evident when the pair-wise difference in the CO_2 storage estimate does not contain zero. The estimates from

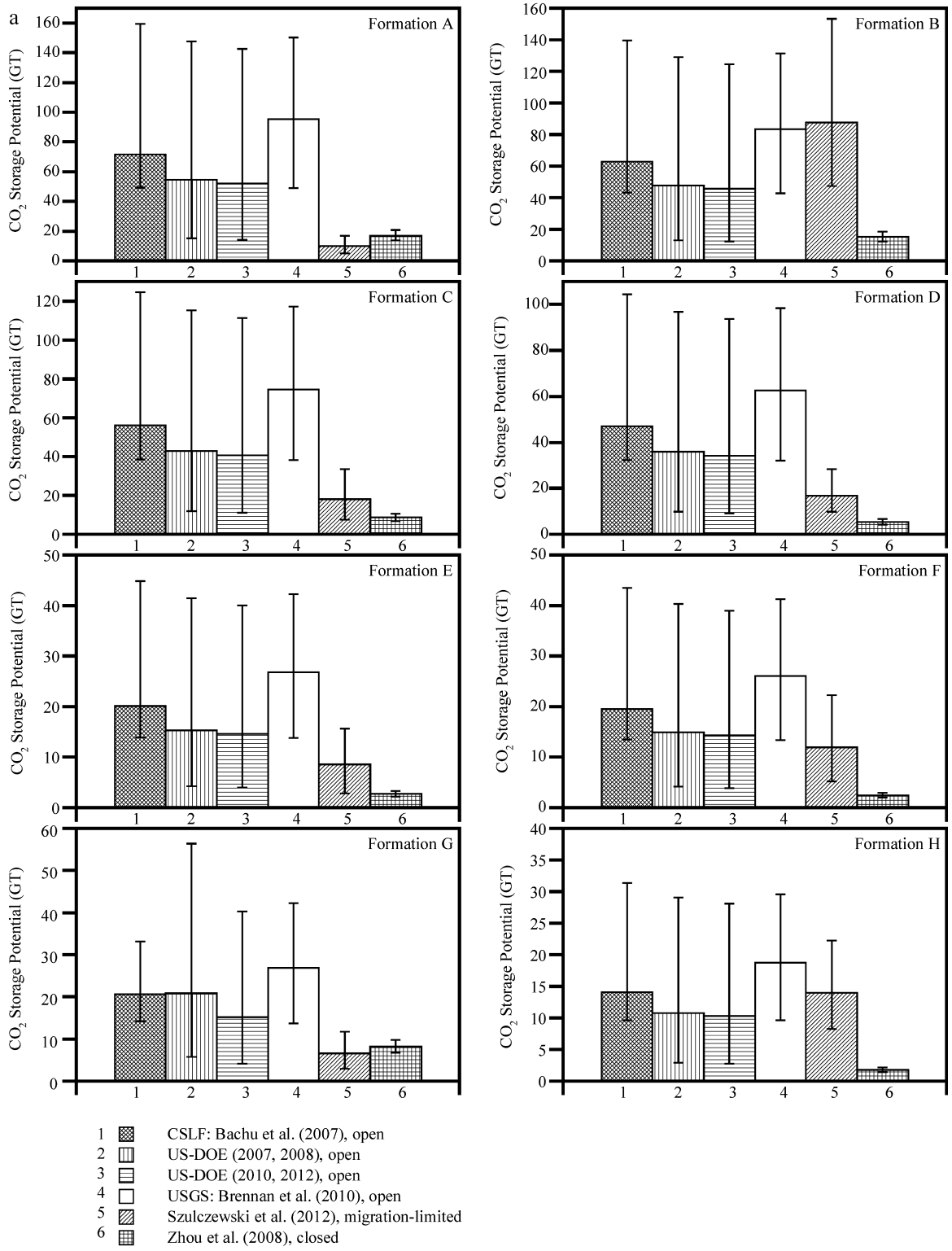


Fig. 1. (a) CO_2 storage resource in gigatonnes (GT) at the 50th percentile for Formations A–H for the following methods: CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), USGS: Brennan et al. (2010), Szulczewski et al. (2012), and Zhou et al. (2008). Black solid error bars represent low and high estimates of CO_2 storage resource based on their 10th and 90th percentiles, respectively. (b) CO_2 storage resource in gigatonnes (GT) at the 50th percentile for Formations I–M for the following methods: CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), USGS: Brennan et al. (2010), Szulczewski et al. (2012), and Zhou et al. (2008). Black solid error bars represent low and high estimates of CO_2 storage resource as represented by their 10th and 90th percentiles, respectively.

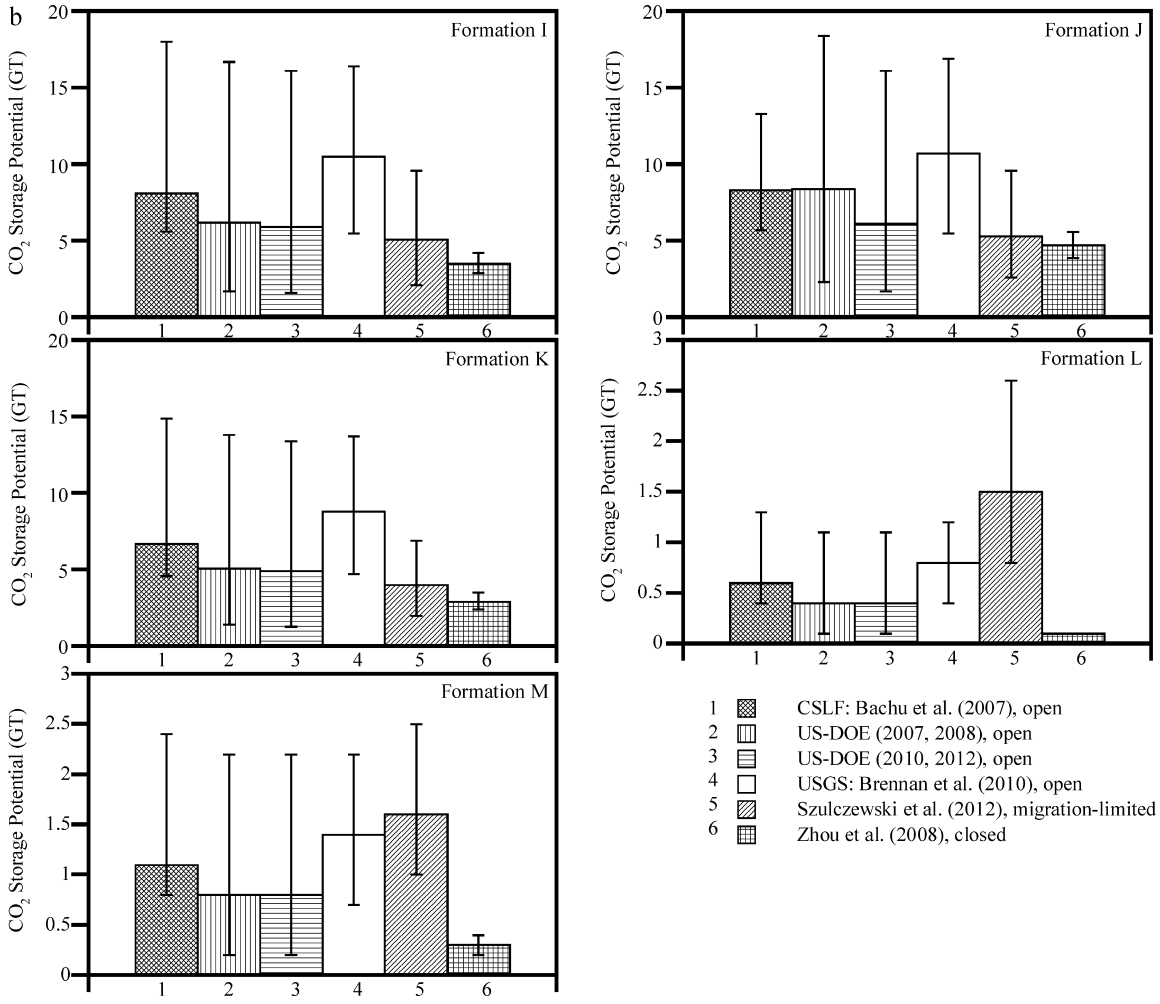


Fig. 1. (*Continued*).

each method were compared in pairs by using the Šidák inequality to control for false positives arising from multiple comparisons (Abdi, 2007). This allowed for fifteen different comparison scenarios, or all possible pairs of six methods. The Šidák inequality is used to compute uncertainty intervals as well as hypothesis testing and is generally applicable for any type of dependence among the parameters studied (Abdi, 2007). The family-wise confidence level for each set of comparisons was set at the 95% confidence level.

Because the CO_2 estimates for most methods were computed using stochastic input values generated by the GoldSim Monte Carlo simulation software, the results from each method were compared using Monte Carlo simulation as well. The probability distribution for each pair-wise difference between methods was simulated by reading the GoldSim simulation output for each method into the program R, "a language and environment for statistical computing and graphics" (R-Project, 2008) and then computing pair-wise differences. Since the GoldSim output for each method is a vector of 50,000 individual simulated values, the result for each pairwise difference is a vector of 50,000 differences. Quantiles from the distribution of these values were then used to determine an uncertainty interval for each comparison using the Šidák inequality to set the confidence level for each individual interval to achieve the desired family-wise confidence level for all comparisons within each formation. After obtaining an approximation to the probability distribution for each difference, the values were smoothed using a kernel density estimation procedure with the R function density. This type of uncertainty analysis, based on the propagation of probability distributions, is a standard approach to uncertainty analysis in the measurement science community. The rationale and techniques for this method are laid out in the *Supplement 1 to the Guide to the Expression of Uncertainty in Measurement* written by the Joint Committee on Guides in Metrology (JCGM, 2008). The R code used to carry out the pair-wise comparison is given in Supplementary Section.

A summary of the pair-wise comparison of storage estimates generated from multiple CO_2 storage methods applied to a uniform saline data set is shown in Fig. 2a and b for formations A–M. Statistically different method pairs – as determined when the difference in the CO_2 storage estimate does not contain zero – are highlighted in gray (Fig. 2a and b). In the case for Formation H, for example, the method pairs USGS: Brennan et al. (2010) and Zhou et al. (2008); CSLF: Bachu et al. (2007) and Zhou et al. (2008); and Szulczewski et al. (2012), and Zhou et al. (2008) were found to be statistically different because their pair-wise comparison difference did not contain zero. In the cases for Formations G and J, no methods were found to be statistically different. Table 4 shows a combined summary of the pair-wise statistical analysis for formations A through M. Statistically different method pairs are identified with crossed boxes (X). The 195 pair-wise distributions generated by the R code for thirteen different saline formation data sets can be found in the *NETL Technical Report Series* (Goodman et al., 2013).

The results of this statistical pair-wise analysis show that the open-boundary methods in general give estimates that do

not statistically differ from one another at the 95% confidence level. Although there are definite differences in the underlying assumptions for open- and closed-boundary methods, CO_2 storage estimates from the closed-boundary method could not be statistically distinguished from those generated by the open-boundary methods for most of the saline formation cases that were studied. In some cases, however, the open-boundary methods do give statistically significant different results when compared to the closed-boundary method. As noted previously and is typical for saline formations, the geologic data describing these formations was sparse due to a deficiency or absence of geologic data. As geologic properties are better defined for potential carbon

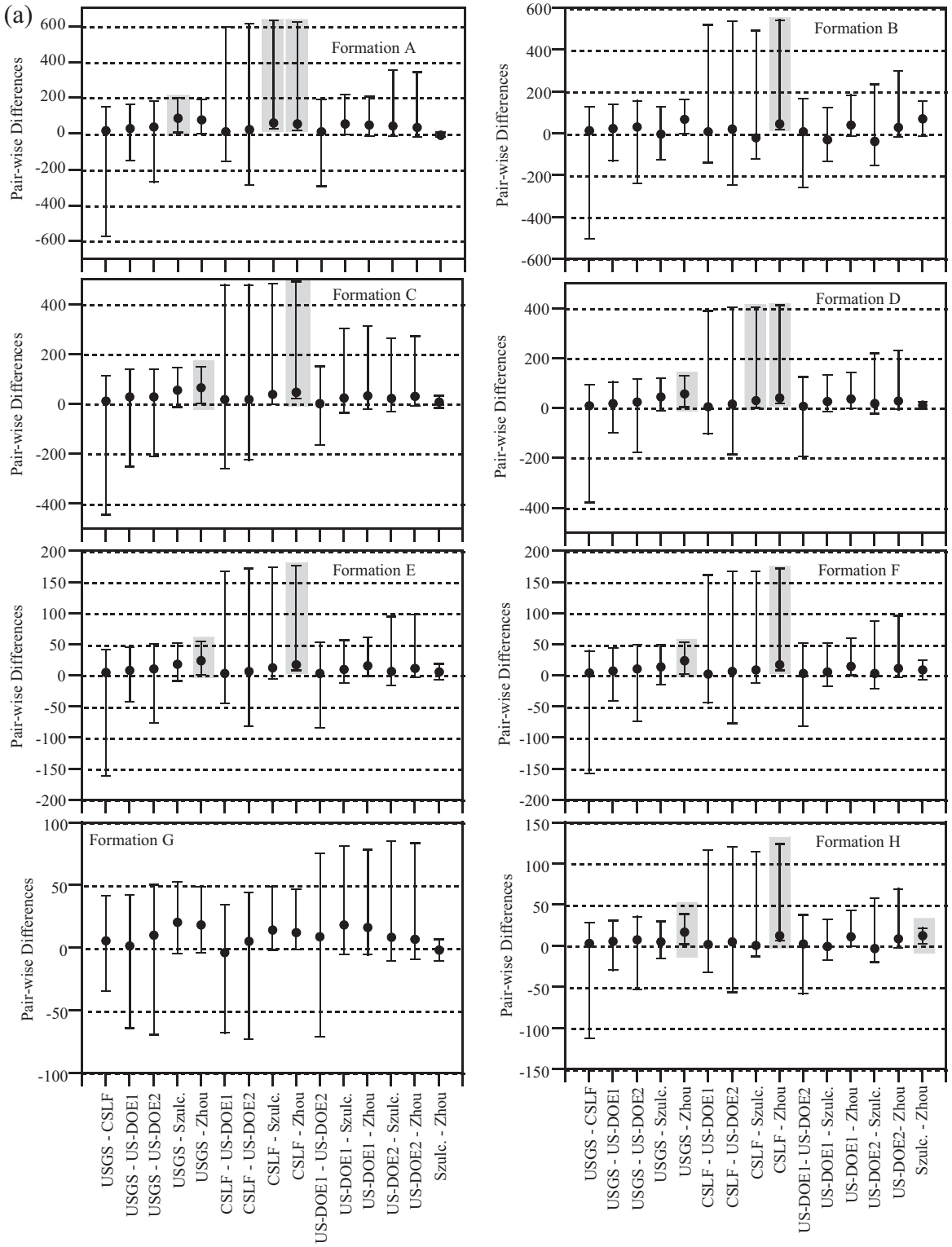


Fig. 2. (a) Simultaneous 95% uncertainty interval for all pair-wise differences between all methods for Formations A–H. Statistically different method pairs are highlighted in gray. (b) Simultaneous 95% uncertainty interval for all pair-wise differences between all methods for Formations I-M. Statistically different method pairs are highlighted in gray.

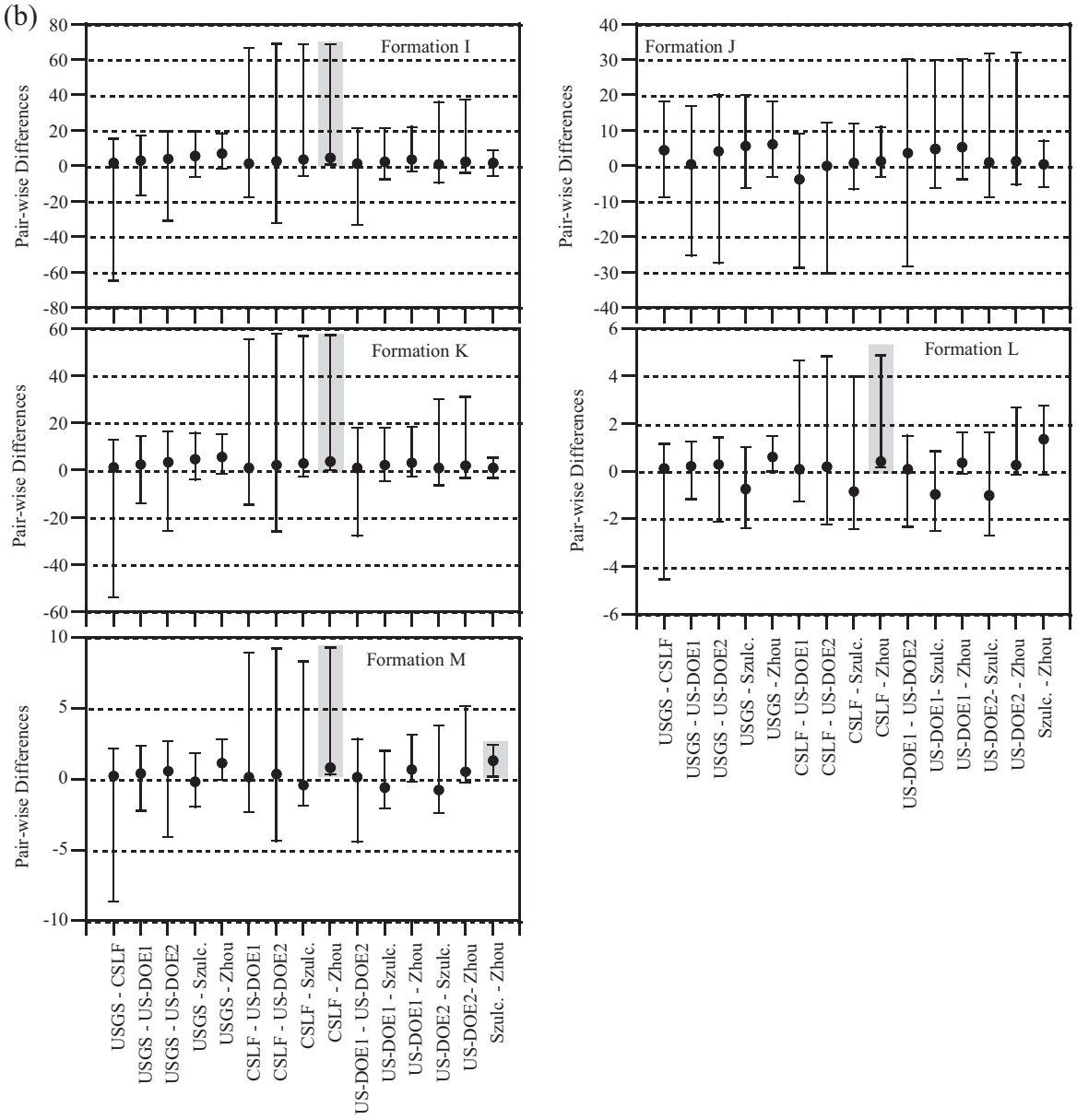


Fig. 2. (*Continued*).

storage formations and uncertainty in underlying geologic parameters decreases, statistical differences between methods may be found.

5.2. Comparison of CO_2 storage estimates with the US-DOE method

A pair-wise analysis based on the propagation of probability distributions of the US-DOE (2010, 2012) method is highlighted to examine differences in CO_2 storage estimated by the US-DOE (2010, 2012) method and each of the other five methods by (a) CSLF: Bachu et al. (2007), (b) US-DOE (2007, 2008), (c) USGS: Brennan et al. (2010), (d) Szulczewski et al. (2012) and (e) Zhou et al. (2008). In all saline formation data set cases, the uncertainty bounds based on these distributions contain zero at the 95% confidence level, demonstrating that the results from the US-DOE (2010, 2012) method do not differ statistically from the results from the other methods for this data set. For brevity, results for formation A are displayed in Fig. 3. The middle red line indicates the median value for each distribution and provides a measure of how well the methods compare. The methods by (a) CSLF: Bachu et al. (2007), (b) US-DOE (2007, 2008), (c) USGS: Brennan et al. (2010), and (d) Szulczewski et al. (2012) compare very well. A complete set of pair-wise probability density distributions of the US-DOE (2010, 2012) method and the other methods can be found in Supplementary Section.

5.3. Comparison of CO_2 storage efficiency

Each method selected in this comparison effort defines and describes the application of CO_2 storage efficiency in different ways. Storage efficiency is a catch-all term that is used to specify the fraction of accessible pore space available to store CO_2, account for boundary conditions, and accommodate CO_2 trapping mechanisms. Comparison of the CO_2 storage efficiency uniquely applied by each method is not useful because of the wide variety of definitions and rules prescribed by the individual methods for applying efficiency. Thus, we define an *Overall Efficiency* that is equal to the total CO_2

Table 4
Summary of all pair-wise differences between methods at the 95% uncertainty interval. The white unfilled boxes represent statistically different method pairs.

	Formation												
Pair-wise Differences	A	B	C	D	E	F	G	H	I	J	K	L	M
USGS - CSLF													
USGS - AtlasI,II													
USGS - AtlasIII,IV													
USGS - Szulc.													
USGS - Zhou													
CSLF - AtlasI,II													
CSLF - AtlasIII,IV													
CSLF - Szulc.													
CSLF - Zhou													
AtlasI,II - AtlasIII,IV													
AtlasI,II - Szulc.													
AtlasI,II - Zhou													
AtlasIII,IV - Szulc.													
AtlasIII,IV - Zhou													
Szulc. - Zhou													

*white boxes represent statistical differences

storage resource estimate predicted by each method divided by the total reservoir pore space available for CO_2 storage specific to each formation as defined in Eq. (9).

$$\text{Overall efficiency} = \frac{\text{Total } CO_2 \text{ storage estimate unique to each method } (GT)}{\text{Total reservoir pore space specific to each formation } (GT)} \quad (9)$$

Under this definition, the *Overall Efficiency* is normalized and CO_2 storage efficiency can be directly compared amongst the six methods. Comparisons of the *Overall Efficiency* for CO_2 storage for the methods are represented in Fig. 4 and Table 5 at the 50th percentile. The *Overall Efficiencies* for the open system methods – CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), and USGS: Brennan et al. (2010) – range between 1.5% and 3.6%; for the closed system method by Zhou et al. (2008) ranges between 0.3% and 1.2%; for the method by Szulczewski et al. (2012), which considers migration-limited storage based on the geology of the formation, ranges between 0.4% and 7.2. Low and high percentiles for *Overall Efficiency* are compiled in the Supplementary Section.

6. Implications of CO_2 storage method choice

Policy makers need dependable CO_2 storage estimates to evaluate and invest in technologies for CCUS. Multiple methods are available to the public that can be applied to estimate CO_2 storage in saline formations. Each method applies its own approach, defines its own terminology, specifies certain mechanisms, includes its own efficiency term, defines its own boundary conditions, and in some cases, its own classification storage scheme. In this study, we compared methods to estimate CO_2 storage in saline formations to determine if the method used for estimation of storage resource significantly impacts the results. We adapted a common saline data set that included thirteen formations to provide uniform geologic inputs required by each assessment method. As is typical with saline formations, the geologic data compiled was sparse. We selected six methods available to the public and applied the methods using the same geologic input parameters to thirteen saline formations to estimate CO_2 storage potential. Both open and closed boundary methods were considered. A rigorous statistical comparison indicated that the open-boundary methods, in almost all cases, do not differ in a statistically significant way in each of the saline formation cases that were studied. A specific comparison of the US-DOE (2010, 2012) method to the other methods demonstrated that the results from the US-DOE (2010, 2012) method do not differ statistically from the results from the other methods for this data set. Although there are definite differences in the underlying assumptions for open- and closed-boundary methods, the estimates of CO_2 storage resource from the closed-boundary method could not be statistically distinguished from those generated by the open-boundary methods for most of the saline formation cases that were studied. In some cases, however, the open-boundary methods do give statistically significantly different results when compared to the closed-boundary method. Uncertainty introduced in storage estimates, resulting from variability and uncertainty in reservoir characteristics, are much larger than any uncertainty in method choice. As more information becomes available in defining reservoirs, uncertainties introduced by choice of method may be more significant.

Assessments of CO_2 storage potential made at the prospective level can be treated as giving comparable results relative to our typical knowledge of the relevant geologic input values when assessing CO_2 storage potential. Because different methods yield similar storage estimates at the prospective level, CO_2 storage method choice is not critical. In general, the uncertainty in the underlying geologic parameters has a much greater impact on overall estimates of CO_2 storage resource than the choice of method does, especially within subsets of methods defined by their boundary assumptions. The closed system estimates were consistently lower than those of the open system methods, but the estimated values from the closed system were also mostly well within the uncertainty bounds of the open system estimates. The fact that the closed methods often give lower results than the open methods suggests that statistically significant differences will ultimately be found between the two methods, but maybe not in all cases, as uncertainty in the underlying geologic parameters decreases. It will be important to

Table 5
Overall Efficiency (%) as defined by Eq. (9)[a].

Formation	CSLF: Bachu et al. (2007) open	US-DOE (2007, 2008) open	US-DOE (2010, 2012) open	USGS: Brennan et al. (2010) open	Szulczewski et al. (2012) migration-limited	Zhou et al. (2008) closed
A	3	2	2	4	0.4	0.6
B	3	2	2	4	4	0.6
C	3	2	2	4	0.9	0.4
D	3	2	2	4	1	0.3
E	3	2	2	4	1	0.4
F	3	2	2	4	2	0.3
G	2	2	1	3	0.6	0.8
H	3	2	2	4	3	0.3
I	3	2	2	4	2	1
J	2	2	1	3	1	1
K	3	2	2	4	2	1
L	3	2	2	4	7	0.7
M	3	2	2	4	4	0.7

[a] Overall efficiency based on 50th percentile.

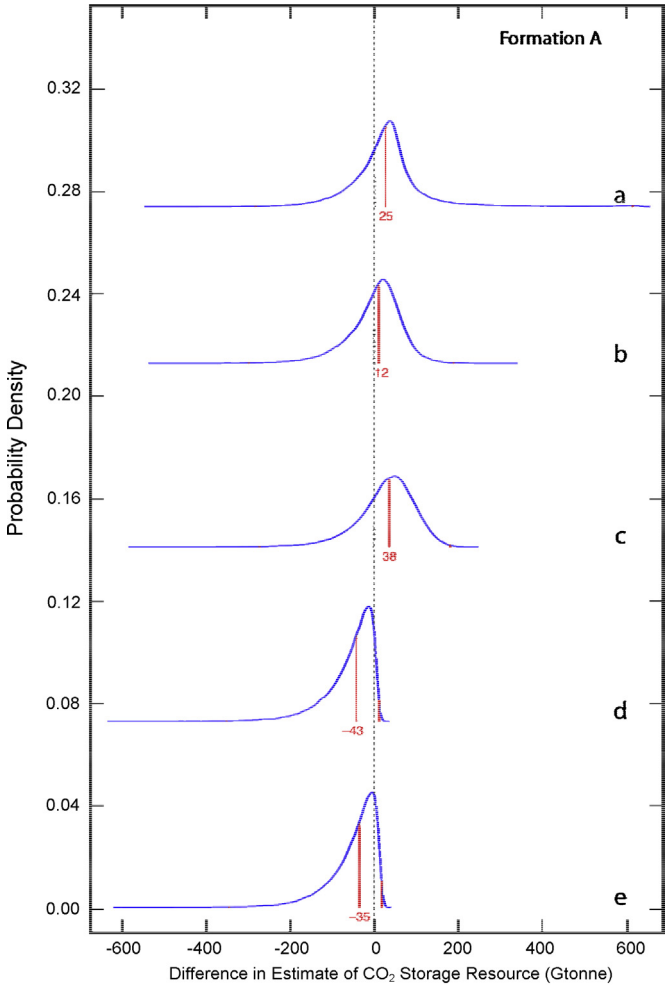


Fig. 3. Comparing the predicted distributions of the pair-wise differences between the method used in the US-DOE (2010, 2012) with (a) CSLF: Bachu et al. (2007), (b) US-DOE (2007, 2008), (c) USGS: Brennan et al. (2010), (d) Szulczewski et al. (2012) and (b) Zhou et al. (2008).

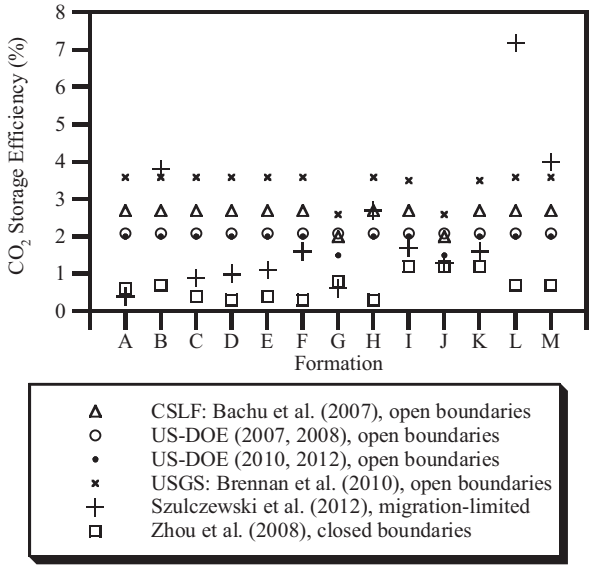


Fig. 4. Overall Efficiency at the 50th percentile, for the following methods: CSLF: Bachu et al. (2007), US-DOE (2007, 2008), US-DOE (2010, 2012), USGS: Brennan et al. (2010), Szulczewski et al. (2012), and Zhou et al. (2008).

determine whether closed or open boundary conditions are appropriate on a site specific basis.

7. Disclaimer

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ijggc.2013.07.016.

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